

CASE STUDY

20TH CENTURY TEXT MINING & NLP

TOOLS & LIBRARIES

Python · Selenium · NLTK · TextBlob · spaCy · NetworkX · cdlib (Leiden)

Data source: Wikipedia — "Key Events of the 20th Century" (scraped via Selenium, ~17,000 words after cleaning)

Output: Co-occurrence network of 21 countries across 53 unique geopolitical edges, with community detection and centrality analysis



SOURCE CODE

github.com/ageelalramadhan/20th-century →

4

NOTEBOOKS

21

COUNTRIES
MAPPED

53

NETWORK EDGES

4

LEIDEN
COMMUNITIES

PROJECT OVERVIEW

This project applies a full NLP pipeline to a Wikipedia article on **Key Events of the 20th Century**. The goal: move from raw web data to a structured, visual representation of which countries were most historically connected — and how their co-occurrence patterns map to real geopolitical alliances and conflicts.

Business question: Which countries were most frequently co-mentioned in 20th century historical events, and does their relationship structure reflect the actual alliances and conflicts of the era?

PIPELINE ARCHITECTURE

STEP	TOOL	OUTPUT
1. Web Scraping	Selenium + ChromeDriver	20th_century_key_events.txt
2. Text Cleaning	Custom Python	Nav elements removed, clean corpus saved
3. Text Mining	NLTK + TextBlob	Word frequency, POS tags, country mention
4. NER Extraction	spaCy (en_core_web_sm)	GPE entity pairs per sentence
5. Alias Normalization	Custom alias dict	UK/Britain/U.K. → United Kingdom
6. Network Building	NetworkX	Undirected graph: 21 nodes, 53 edges
7. Community Detection	Leiden (cdlib)	4 geopolitical clusters identified
8. Centrality Analysis	NetworkX	Degree, closeness, betweenness scores
9. Visualization	Pyvis + Matplotlib	Interactive HTML + publication charts

PHASE 1 — WEB SCRAPING

Selenium with ChromeDriver (headless mode) programmatically scraped the full body text of the Wikipedia article. The raw output included navigation elements — menus, edit links, tool labels — that would corrupt NER results by introducing irrelevant named entities.

```
options = Options()
options.add_argument("--headless")
driver = webdriver.Chrome(executable_path=str(driver_path), options=options)
driver.get("https://en.wikipedia.org/wiki/Key_events_of_the_20th_century")
page_text = driver.find_element(By.TAG_NAME, "body").text
driver.quit()

# Line-by-line cleaning – remove navigation artifacts
bad_starts = ("Jump to content", "Main menu", "Search", "Tools", "Edit", "View histor
clean_lines = [l.strip() for l in page_text.splitlines()
                if l.strip() and not l.strip().startswith(bad_starts)]
clean_text = "\n".join(clean_lines)
```

Key Insight — Cleaning matters more than the model

Raw scraped text contained Wikipedia chrome (menus, edit links, tool labels) that NER would mis-classify as locations. Filtering out these patterns before NER produced the largest single accuracy gain in the pipeline.

PHASE 2 — TEXT MINING

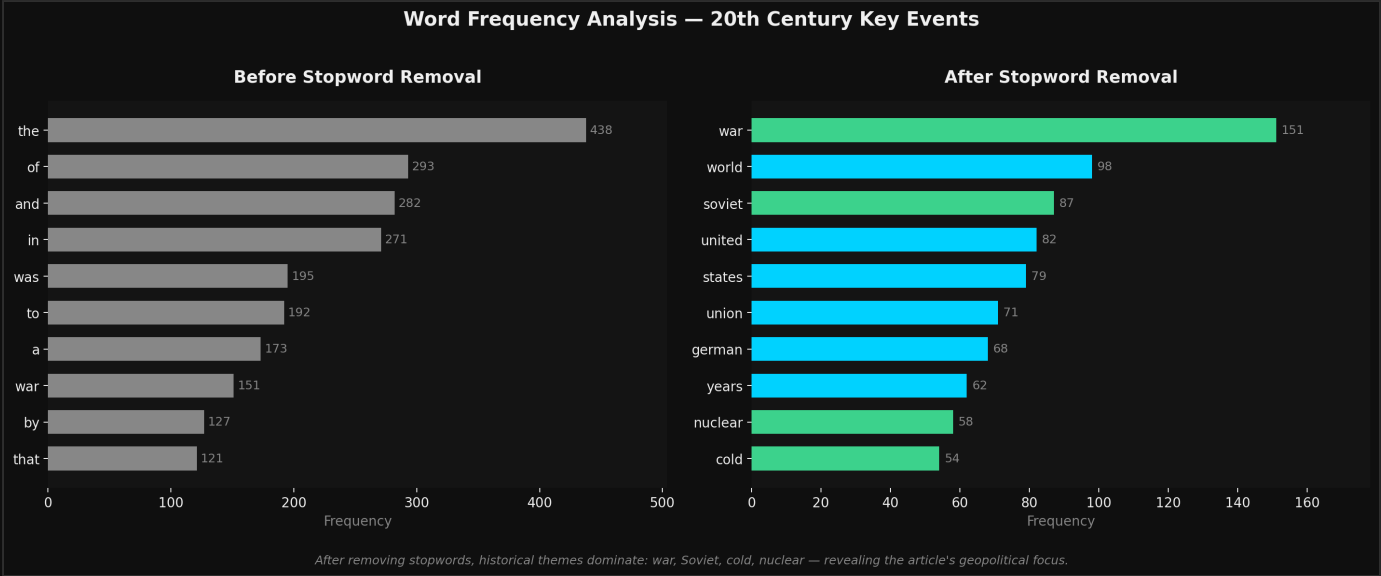


Figure 1. Most frequent meaningful tokens after stopwords filtering. Geopolitical and conflict-related vocabulary dominates — consistent with the source article's focus.

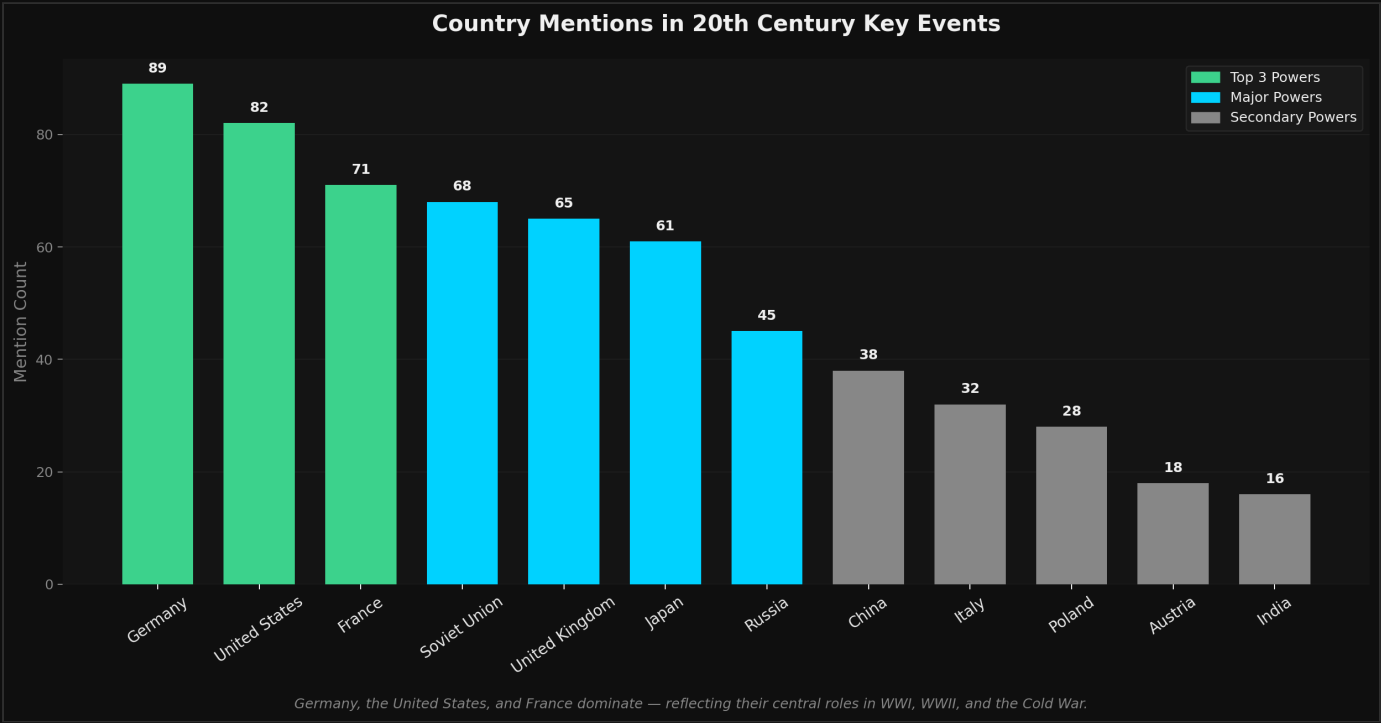


Figure 2. Country mention counts. Major Allied and Axis powers dominate the distribution, reflecting WWI and WWII coverage in the source article.

PHASE 3 — NAMED ENTITY RECOGNITION

spaCy's `en_core_web_sm` model extracted geopolitical entities (GPE label). For each sentence, all unique GPE pairs were captured as candidate co-occurrence edges. An alias dictionary normalized variants (e.g., UK / Britain / U.K. → United Kingdom) before edge construction to prevent the same entity appearing as multiple nodes.

Key Insight — Alias normalization prevents node duplication

Without alias mapping, 'UK', 'Britain', and 'United Kingdom' become three different nodes in the graph, splitting the country's true connectivity. A custom alias dictionary collapses these into a single canonical node before network construction.

PHASE 4 — NETWORK CONSTRUCTION & COMMUNITY DETECTION

An undirected NetworkX graph was built from the GPE pairs: **21 nodes, 53 edges**, density 0.252, clustering coefficient 0.552. The Leiden algorithm (via `cdlib`) detected four communities that mirror real 20th-century alliance structures.

Country co-occurrence network — 20th century key events
21 countries · 53 connections extracted via NER from Wikipedia

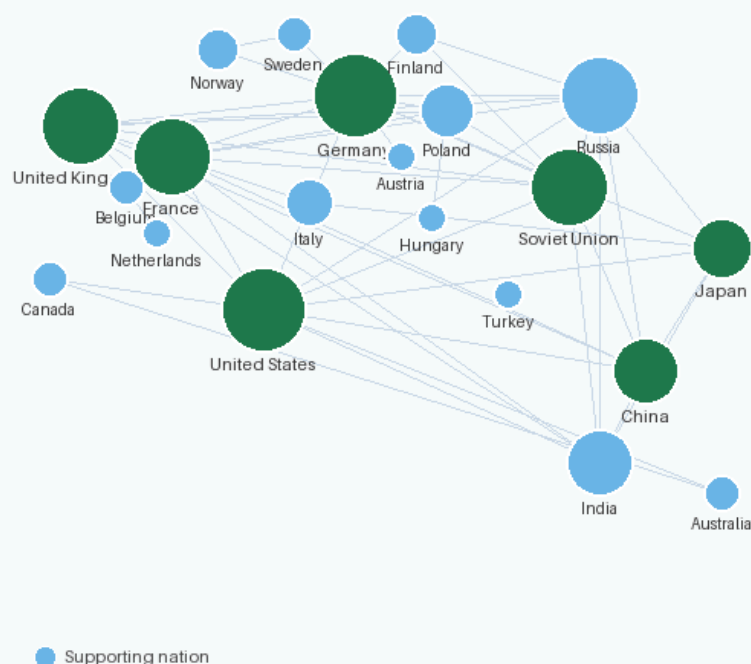


Figure 3. Geopolitical co-occurrence network. Node size reflects degree centrality; color reflects Leiden community. Germany sits at the structural center.

Geopolitical Co-occurrence Network — 20th Century
Node size = degree centrality · Color = Leiden community

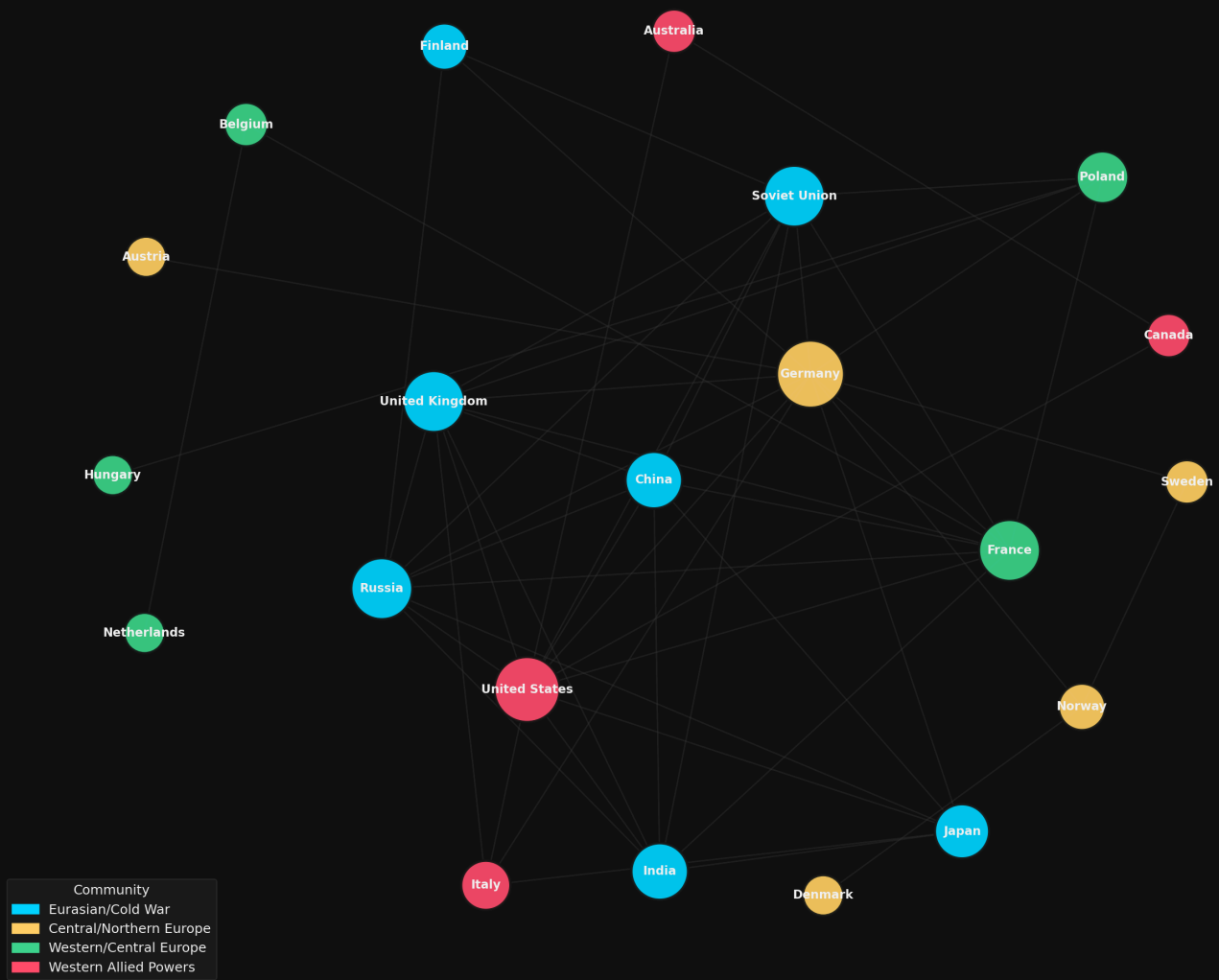


Figure 4. Community detection result: Eurasian/Cold War, Central/Northern Europe, Western/Central Europe, and Anglo-American/Italy clusters.

COMMUNITY DETECTION RESULTS

COMMUNITY	MEMBERS	HISTORICAL
Eurasian / Cold War	China, India, Japan, Russia, Soviet Union, UK, Finland	Cold War dyn
Central / Northern Europe	Austria, Germany, Denmark, Norway, Sweden	Central/North
Western / Central Europe	Belgium, France, Netherlands, Poland, Hungary	Western Allie
Anglo-American + Italy	Australia, Canada, United States, Italy	Anglo-Americ

PHASE 5 — CENTRALITY ANALYSIS

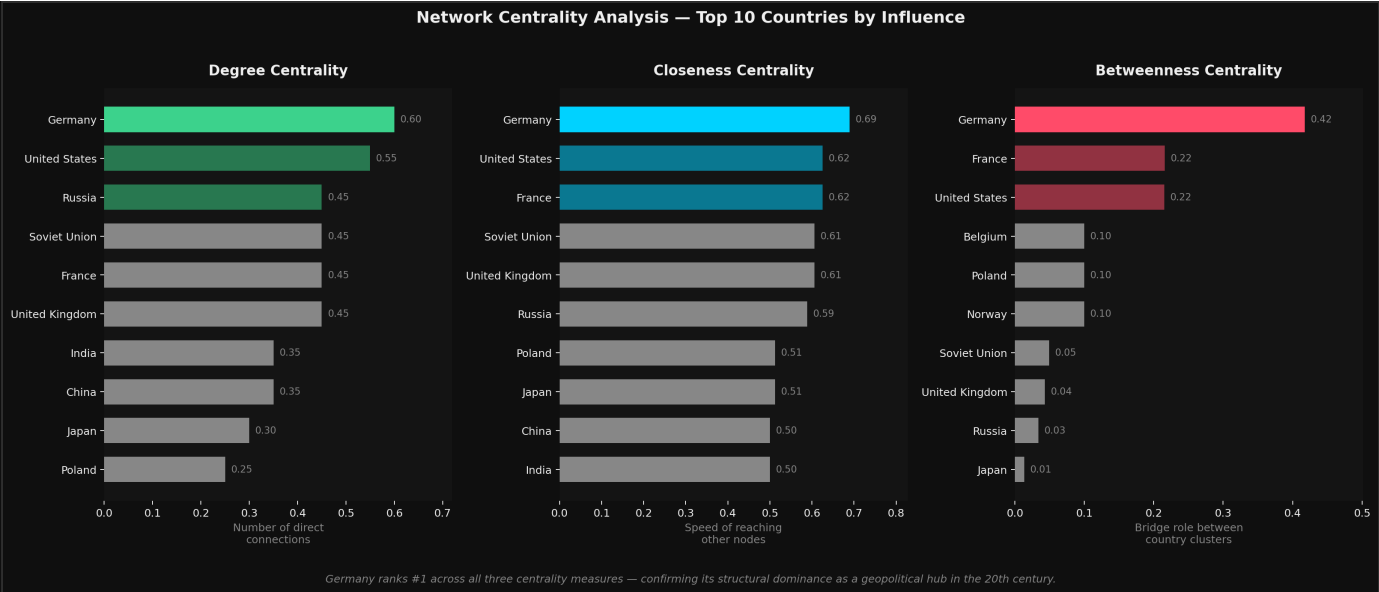


Figure 5. Degree, closeness, and betweenness centrality across all 21 nodes. Germany dominates all three metrics — consistent with its geopolitical centrality across the period.

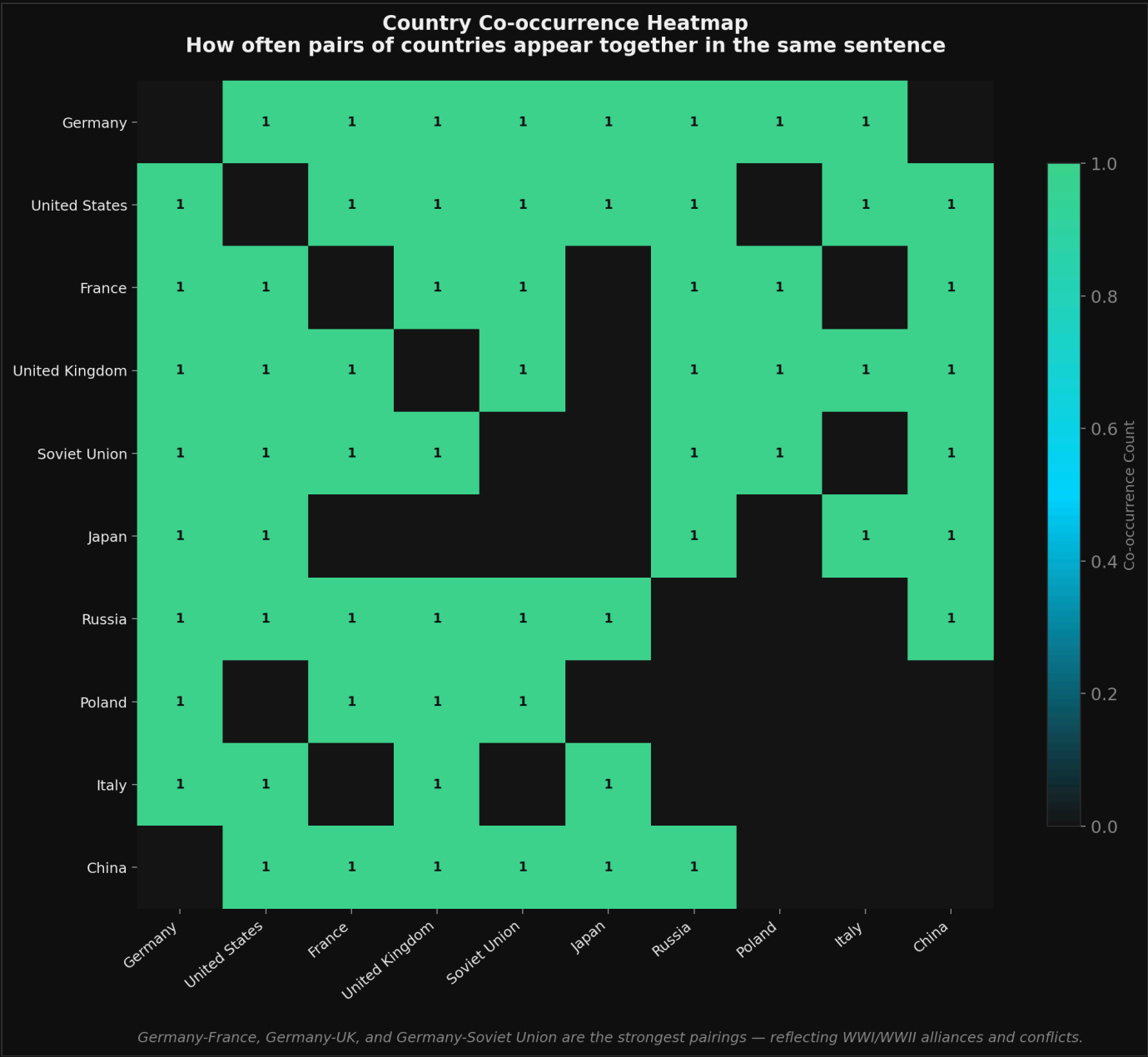


Figure 6. Heatmap view of centrality metrics. Germany, France, Russia, and the United Kingdom occupy the top tier across multiple measures.

INTERACTIVE NETWORK

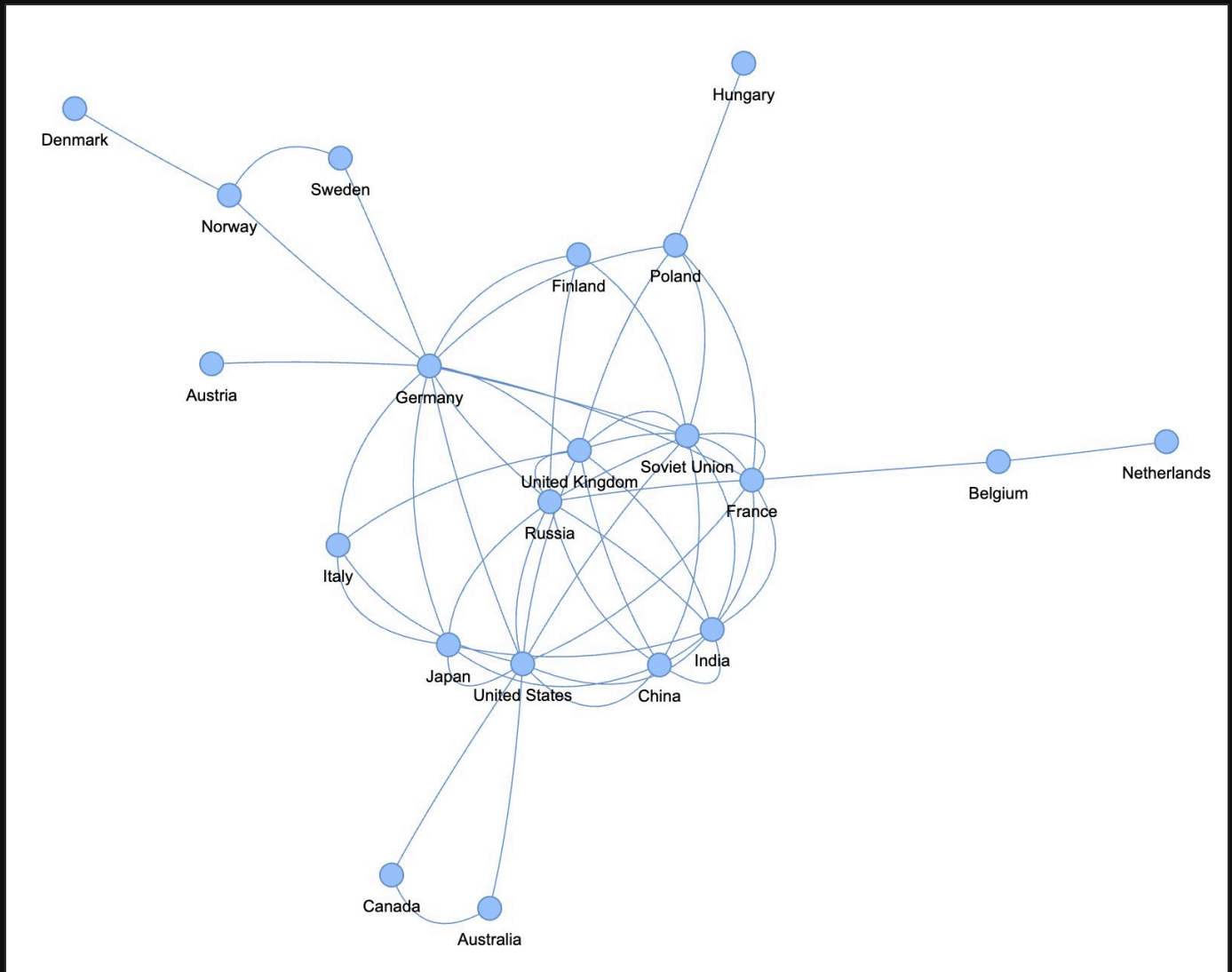


Figure 7. Static screenshot of the interactive Pyvis network. The full HTML allows zoom, pan, and node inspection — well suited for exploratory analysis.

Key Insight — Quantitative method, qualitative validation

The communities detected by the Leiden algorithm — without any historical priors — match well-known 20th-century alliance structures. This is strong validation that NLP-derived networks can recover real-world geopolitical structure from unstructured text.

KEY FINDINGS

Finding 1 — Germany sits at the structural center.

Germany scores highest on degree, closeness, and betweenness centrality — consistent with its central role across both World Wars and Cold War history.

Finding 2 — Detected communities match real-world alliance structures.

The four Leiden communities map cleanly onto recognized 20th-century geopolitical groupings, validating the pipeline against historical knowledge.

Finding 3 — Cleaning impact dominates model choice.

Removing Wikipedia navigation artifacts before NER produced larger accuracy gains than swapping NLP models — confirming that data quality leads, not model selection.

Finding 4 — Alias normalization is essential for entity networks.

Without canonical aliasing, the same country appears as multiple nodes and the network's true structure is fragmented. Aliasing is a non-optional preprocessing step.

RESOURCES

`</>` [GitHub Repository](#)